Yifan Yang

Incoming Ph.D. Student at Southeast University | AI for Software Engineering

Shenzhen • yifanyang6@link.cuhk.edu.cn • yifanyang6.github.io • YifanYang6 • 0009-0008-3952-9207 • Google Scholar

Summary

Incoming Ph.D. student at Southeast University, to be advised by Prof. Tao Xie starting in September 2027. Working on AI for Software Engineering and Coding Agent Runtime optimization. Previously a research assistant advised by Prof. Pinjia He, with research on microservice diagnostics, observability, fault injection, and LLM-based RCA agents.

Education

Ph.D. **Southeast University** — Incoming, Sep 2027
Computer Science and Technology — Nanjing
- Advised by Prof. Tao Xie.

M.Sc. **The Chinese University of Hong Kong, Shenzhen** — Sep 2024 – Present
Artificial Intelligence and Robotics — Shenzhen
- School of Data Science.

B.Mgt. **Shandong University** — Sep 2019 – Jun 2023
Industrial Engineering — Jinan
- Training in systems thinking, optimization, and engineering practice.

Experience

Research Assistant — Aug 2024 – Nov 2026
The Chinese University of Hong Kong, Shenzhen — Shenzhen
- Work under the supervision of Prof. Pinjia He at the School of Data Science.
- Conduct research in AI for Software Engineering and AIOps, with an emphasis on microservice diagnostics, root cause analysis, observability, and fault-propagation-aware evaluation.
- Contribute to end-to-end research workflows across multiple projects, including system implementation, telemetry and data collection, fault injection, system integration, baseline reproduction, and performance evaluation.
- Help maintain lab computing and networking infrastructure, including CPU clusters, GPU resources, Kubernetes environments, PVE virtualization, and OpenWrt-based networking.
- Designed fault-injection campaigns, improved observability integration across microservice systems, contributed to dataset organization, and wrote related sections for OpenRCA 2.0 and RCA benchmark research.
- First author of Gleaner, published at ISSTA'26.

Publications

Gleaner: A Semantically-Rich and Efficient Online Sampler for Microservice Diagnostics — Oct 2026
Yifan Yang, Aoyang Fang, Songhan Zhang, Pinjia He*
ISSTA'26 [PDF][DOI][code]

Beyond Fault Localization: A Trajectory-Level Study of LLM Agents for Microservice Root Cause Analysis — Aug 2026
Qisheng Lu, Aoyang Fang, Junjielong Xu, Jin'ao Shang, Songhan Zhang, **Yifan Yang**, Xiaochuan Yan, Pinjia He*
arXiv preprint [PDF]

Rethinking the Evaluation of Microservice RCA with a Fault Propagation-Aware Benchmark — June 2026
Aoyang Fang, Songhan Zhang, **Yifan Yang**, Haotong Wu, Junjielong Xu, Xuyang Wang, Rui Wang, Manyi Wang, Qisheng Lu, Pinjia He*
FSE'26 [PDF][DOI]

OpenRCA 2.0: From Outcome Labels to Causal Process Supervision — June 2026
Aoyang Fang, **Yifan Yang**, Jin'ao Shang, Qisheng Lu, Junjielung Xu, Rui Wang, Songhan Zhang, Yuzhong Zhang, Boxi Yu, Pinjia He*
arXiv preprint [PDF]

DynaCausal: Dynamic Causality-Aware Root Cause Analysis for Distributed Microservices — Oct 2025
Songhan Zhang, Aoyang Fang, **Yifan Yang**, Ruiyi Cheng, Xiaoying Tang, Pinjia He*
arXiv preprint [PDF]

Awards

First-Class Academic Merit Scholarship — CUHK-Shenzhen MAIR, 1 September 2026.

Service

- Artifact Evaluation Program Committee, ASPLOS 2027
- Artifact Evaluation Program Committee, ACM SOSP 2026
- Artifact Evaluation Program Committee, ACM CCS 2026
- Artifact Evaluation Program Committee, NDSS 2027

Projects

Gleaner — Diagnosis-oriented online trace sampling for microservice observability.
[GitHub]

chaos-experiment — Fault injection tooling for controlled microservice RCA evaluation.
[GitHub]

Aegis — Experiment tooling and systems engineering workspace for AIOps research.
[GitHub]

Skills

Research Areas: Intelligent Software Engineering (AI4SE), AIOps, intelligent observability, LLM-based and multi-modal RCA, fault injection benchmarks
Systems: Linux, Docker, Kubernetes, Helm, Skaffold, OpenTelemetry, Jaeger, PVE, OpenWrt
AI-assisted Development: Codex, Claude Code, Git, LaTeX, research automation workflows
ML and LLM: PyTorch, vLLM, LLM fine-tuning, inference optimization
Languages: Chinese (Native), English (IELTS 6.5)